SYNAPSE EDGE INTELLIGENCE

WAF Rule Pipeline

248 production rules across 30 threat classes — evaluated in two phases with a single canonical 403 exit. Deferred rules see what the body-phase pass can't.

248	30	2	403
PRODUCTION RULES	THREAT CLASSES	EVAL PHASES	CANONICAL BLOCK

EVALUATION PIPELINE — BODY PHASE · DEFERRED PHASE · VERDICT

STAGE 01
Index Candidates
RuleIndex narrows 248 rules to the candidate subset for this request via cached keys & regex groups.
PREFILTER

STAGE 02
Body-Phase Pass
Evaluates all rules except those whose match kinds need data not yet available (dlp_violation).
MAIN PASS

STAGE 03
DLP Scanner
Async body scan produces dlp_violation findings. Gates the deferred pass.
ENRICH

STAGE 04
Deferred Pass
Re-evaluates only deferred_rule_indices — the rules skipped in stage 02 — now with DLP data in scope.
POST-DLP

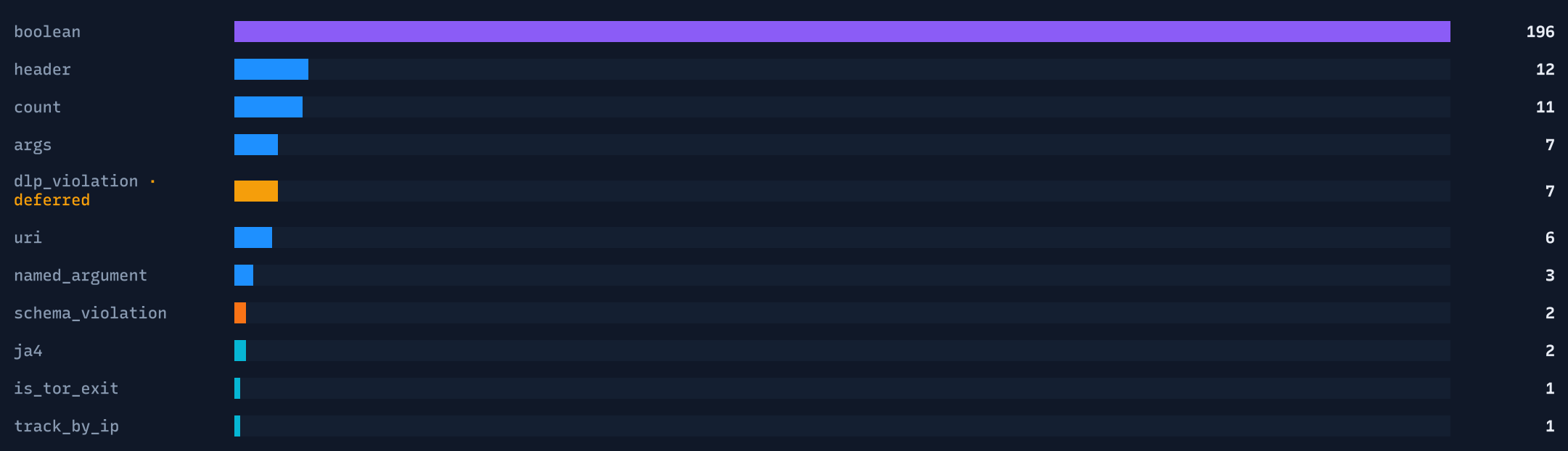
STAGE 05
Verdict → 403
Any blocking match exits through the single canonical WAF block response. Non-blocking matches add risk.
EMIT

> **Why two phases?**
> Some rules (dlp_violation match kind) need scanner output that isn't available during the synchronous body pass. Rather than blocking the pipeline, those rule indices are tracked in deferred_rule_indices, skipped via an O(1) HashSet in the body pass, and re-evaluated once DLP completes — so the main pipeline stays fast.

TOP THREAT CLASSIFICATIONS — 248 RULES ACROSS 30 CLASSES

Known Vulnerability 37	Toolkit 35	Command Injection 19	Programmatic Access 17
Bad Bot 14	Web Attack 13	SQL Injection 12	Directory Traversal 11
Info Disclosure 11	Evasion 9	Software Detection 9	Bad Traffic 8
XSS 6	Content Enumeration 6	Botnet Activity 5	+ 15 more classes 26

HOW RULES MATCH — MATCH KIND DISTRIBUTION



BLOCKING VS OBSERVATIONAL RULES

62
BLOCKING RULES · 25%
Trigger an immediate canonical 403. Reserved for high-confidence signatures: known exploits, confirmed toolkits, active injection patterns.

186
OBSERVATIONAL RULES · 75%
Contribute to the actor risk score without blocking directly. Feeds the risk lifecycle — repeated low-confidence hits accumulate into challenge or block decisions.

CANONICAL BLOCK RESPONSE — ONE EXIT POINT

RESPONSE BODY (STABLE)

```
{"error": "access_denied"}
```

GUARANTEE

Byte-identical output from every block site — body-phase, deferred, or future additions — via send_waf_block_response. Clients that parse the shape depend on it; treated as an API contract.

RESPONSE HEADERS

:status	403
content-type	application/json
content-length	22
x-request-id	<echoed>
strict-transport-security	<https only>
+ security headers	X-Frame, CSP, …

Two phases. One exit. 248 rules pulling in the same direction.
Fast-path body-phase evaluation with a deferred pass for rules that need DLP output. Every block emits the same canonical 403.